SAI SRUTHI RENATI

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Linkedin | Website

Education

University of Texas at Arlington (UTA), Texas Aug 2024 – Dec 2026
Master of Science (MS) in Computer Science *GPA: 3.83*
Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Python, Data Mining, Data Analysis & Modeling Techniques, Design & Analysis Algorithms, Linux, Advanced Software Testing

NBKR Institute of science and Technology, India Aug 2018 – Jul 2022
Bachelor of Technology (B.Tech.) in ECE *GPA: 3.45*

Professional Experience

Application Developer Sep 2022 – July 2024
IBM *Chennai, India*
- Maintained and enhanced a 100,000+ line **Mainframe** codebase using COBOL, JCL, SonarScanner, and SonarQube for a high-stakes US-based project.
- Delivered 10+ features and resolved 95% of user issues within 24 hours, achieving 98% satisfaction and completing tasks 20% ahead of schedule.
- Consistently earned 'A' grades in all Software Build Reviews (SBRs) and was recognized as an **"Emerging Rockstar"** for high-quality, efficient delivery.

Data Scientist (Intern) May 2021 – June 2021
VI Solutions *Bengaluru, India*
- Developed an end-to-end **"smart health monitoring system"** using Python and TensorFlow, achieving 92% model accuracy in predicting anomalies from sensor data using an LSTM neural network.
- Integrated real-time IoT sensor streams with the LSTM model, reducing health alert response time by 40%.

Projects

Electric Vehicle Analysis Dashboard *—Power BI, DAX, Data Modeling, Power Query*
- Built an interactive dashboard that analyzes the adoption of electric vehicles, range trends, and state distribution with slicer-driven interactivity 100%.
- Implemented advanced DAX measures and data modeling to calculate electric range evolution, BEV vs PHEV share, and CAFV eligibility segmentation, improving analytical clarity and visual storytelling by 80%.

Skillsync- Matching you with the right job *—React JS, Node JS, My SQL, Selenium, PyTest*
- Reduced back-end data transformation latency by 35% through memory-efficient query handling and parallel processing using optimized Python services.
- Integrated Bazel and PyTest for continuous integration pipelines, reducing deployment errors by 25% and improving testing turnaround time..

Twitter Data Analysis using SVM,RNN-BiLSTM, DistilBERT-BiLSTM *— Python, Keras, Pytorch, Matplotlib*
- Enhanced transformer model inference speed by 35% through GPU-accelerated kernel optimization using PyTorch and applied performance tuning via TorchScript and Nvidia Nsight.
- Reduced memory footprint by 30% and improved inference efficiency by integrating caching strategies and optimizing data pre-processing pipelines.

Movie Search Engine using FAISS and Embeddings *— Python, FAISS, Hugging Face Transformers, Scikit-learn*
- Reduced search latency by 60% by implementing optimized FAISS indexing with memory-efficient similarity search, utilizing SIMD and multithreading techniques.
- Improved efficiency of the retrieval pipeline by 40% using caching layers and optimizing the memory hierarchy for faster retrieval and classification of embedding.

Heart Disease Prediction using Machine Learning Models *— Python, SciPy, Seaborn, matplotlib*
- Developed a heart disease prediction pipeline using 5 ML models, achieving 91% accuracy through end-to-end data cleaning, 100% preprocessing coverage, and model evaluation with confusion matrix.

Skills

Programming Languages: Python, PySpark, My SQL, C, C++, Java, COBOL, JCL
Scripting Languages: HTML, CSS, JavaScript, React.Js, Flask
Data Science Libraries: NumPy, Pandas, Scikit-learn, Streamlit, Selenium, Matplotlib, Seaborn
Machine Learning: Classification, Regression, Clustering, Reinforcement Learning, Model Evaluation
Deep Learning: ANN, CNN, RNN-BiLSTM, CNN-BERT, Transformers, GANs, PyTorch, TensorFlow, LLMs
NLP & GenAI: Hugging Face, Tokenization, Prompt Engineering, Retrieval-Augmented Generation (RAG)
Data Visualization: Matplotlib, Seaborn, Power BI
Cloud & Tools: AWS, SonarQube, Xpediter, Cypress, Jest, A/B Testing, Git, Docker